CIC Block Mathematical Analysis

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General Structure of a CIC and Integrator Section

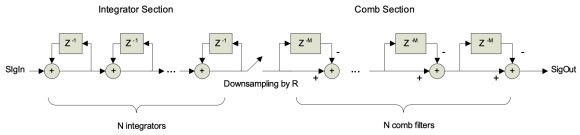


Figure 1: CIC schematic.

The signal $x[n]$ passes through a chain of delays z^{-1} with adders, followed by a **downsampling** block, producing the output $y[n]$ that enter in the Comb section. The Integrator section follow this relationship:

$$y[n] = x[n] + y[n-1]$$

Z-Transform of the Integrator

Applying the $\mathcal{Z}$-transform:

- A one-sample delay $(n-1)$ becomes a multiplication by z^{-1}.
- In the $\mathcal{Z}$ domain: $z = Ae^{j\theta}$, therefore $x[n] \overset{\mathcal{Z}}{\Longrightarrow} X(z)$.
- $x[n-k]$, with $k > 0$ and $x[n] = 0$ for $n < 0$, corresponds to $z^{-k}X(z)$.

Starting from $y[n] = x[n] + y[n-1]$, applying $\mathcal{Z}$:

$$Y(z) = X(z) + z^{-1}Y(z)$$

$$Y(z) - z^{-1}Y(z) = X(z)$$

$$Y(z)\left(1 - z^{-1}\right) = X(z) \implies \frac{Y(z)}{X(z)} = \frac{1}{1 - z^{-1}}$$

The **integrator transfer function** is therefore:

$$\boxed{H(z) = \frac{1}{1 - z^{-1}}}$$

CIC with $N = 2$ integrators

Our CIC filter is set with $N = 2$, i.e. 2 integrators in cascade.

0.1 Single Integrator

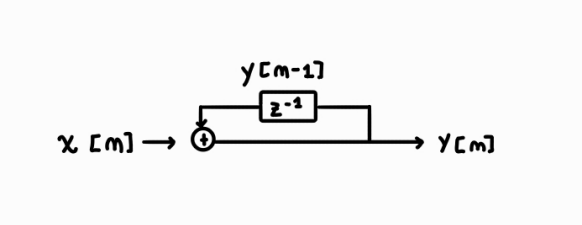


Figure 2: First integrator stage.

$$y[n] = x[n] + y[n-1]$$

0.2 Two Integrators ($N = 2$)

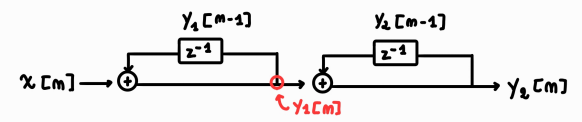


Figure 3: Two integrator stages in cascade.

$$y_1[n] = x[n] + y_1[n-1]$$
$$y_2[n] = y_1[n] + y_2[n-1]$$

The z^{-1} block acts as a *memory register* to store the previous sum and perform the new sum.

1 Transfer Functions of the two integrator stages in cascade

The total transfer function is:

$$H(z)\big|_{N=2} = H_1(z) \cdot H_2(z)$$

Computation of $H_1(z)$

$$y_1[n] = x[n] + y_1[n-1] \overset{\mathcal{Z}}{\Longrightarrow} Y_1(z) = X(z) + z^{-1}Y_1(z)$$

$$H_1(z) = \frac{Y_1(z)}{X(z)} = \frac{1}{1 - z^{-1}}$$

Computation of $H_2(z)$

$$y_2[n] = y_1[n] + y_2[n-1] \overset{\mathcal{Z}}{\Longrightarrow} Y_2(z) = Y_1(z) + z^{-1}Y_2(z)$$

$$H_2(z) = \frac{Y_2(z)}{Y_1(z)} = \frac{1}{1 - z^{-1}}$$

Total Transfer function of the 2-Stage integrator

$$\boxed{H_{\text{tot}}(z) = \left(\frac{1}{1 - z^{-1}}\right)^2}$$

The square exponent justifies the presence of 2 stages. The first integrator **attenuates high-frequency noise** by eliminating fast transitions ($1 \to 0$ or $0 \to 1$) arriving from the microphone. The frequency at which bits arrive is $f = 2{,}4\,\text{MHz}$.

2 Numerical example

Note: PDM bits arrive in $\mathcal{Z}$ with encoding $1 \to +1$, $0 \to -1$. The following table shows the evolution of the two integrators for a sample PDM input:

$x[n]$	$y_1[n-1]$	$y_1[n]$	$y_1[n]$	$y_2[n-1]$	$y_2[n]$
INIT	–	0	–	0	–
1	0	1	1	0	1
1	1	2	2	1	3
0	2	1	1	3	4
1	1	2	2	4	6
1	2	3	3	6	9
1	3	4	4	9	13
0	$4-1$	3			
1	$1+3$	4			
1	$4+1$	5			
1	$5+1$	6			

PDM sequence: 1 1 0 1 1 1 0 1 1 1

$$\xrightarrow{\text{I integrator}} 1\,2\,1\,2\,3\,4\,3\,4\,5\,6 \xrightarrow{\text{II integrator}} 1\,3\,4\,6\,9\,13\,\cdots \longrightarrow 1\,3\,4\,6\,9\,13\,\cdots$$

Summary

- The **first integrator** suppresses quantization noise at high frequencies, transforming rapid $0 \leftrightarrow 1$ transitions into a slower progression that follows the audio signal's behavior (*ramp*).
- The **second integrator** transforms the ramp into a smoother curve (*parabola*), making the audio even less sensitive to continuous $1 \to 0 \to 1$ transitions that represent noise.

3 COMB section

The complete CIC block consists of:

- **Integrator Section** (sampled at $f_s = 2{,}4\,\text{MHz}$)
- **Downsampling** (with factor R, output frequency $f = f_s/R = 2{,}4\,\text{MHz}/50 = 48\,\text{kHz}$)
- **COMB Section** (sampled at $f_{\text{out}} = 48\,\text{kHz}$)

. The output of the COMB section is:

$$y[n] = y_2[Kn] - y_2[Kn - RM]$$

with sampling time $t = KT$, $K \in \mathbb{N}$ and $T = \frac{1}{48\,\text{kHz}}$.
In the $\mathcal{N}$ domain:

$$Y(z) = X(z) - z^{-MR}X(z) = X(z)\left(1 - z^{-MR}\right)$$

Therefore the transfer function of a single COMB stage is:

$$\boxed{H(z) = \frac{Y(z)}{X(z)} = 1 - z^{-MR}}$$

4 Frequency response of the integrator (Frequency domain)

We compute the transfer function in the frequency domain for a single integrator stage. With $z = e^{j\omega}$:

$$H(z) = \frac{1}{1 - z^{-1}} \implies H(\omega) = \frac{1}{1 - e^{-j\omega}}$$

We expand the denominator:

$$1 - e^{-j\omega} = 1 - \cos(\omega) + j\sin(\omega)$$

Using trigonometric identities:

$$\sin\left(\frac{\alpha}{2}\right) = \sqrt{\frac{1 - \cos\alpha}{2}} \quad \Rightarrow \quad \sin^2\left(\frac{\omega}{2}\right) = \frac{1 - \cos(\omega)}{2} \quad \Rightarrow \quad 2\sin^2\left(\frac{\omega}{2}\right) = 1 - \cos(\omega)$$

$$\sin(2\alpha) = 2\sin(\alpha)\cos(\alpha) \quad \Rightarrow \quad \sin(\omega) = 2\sin\left(\frac{\omega}{2}\right)\cos\left(\frac{\omega}{2}\right)$$

Therefore:

$$H(\omega) = \frac{1}{2\sin^2\left(\frac{\omega}{2}\right) + j\,2\sin\left(\frac{\omega}{2}\right)\cos\left(\frac{\omega}{2}\right)} = \frac{1}{2\sin\left(\frac{\omega}{2}\right)\left(\sin\left(\frac{\omega}{2}\right) + j\cos\left(\frac{\omega}{2}\right)\right)}$$

Magnitude of $H(\omega)$

$$|H(\omega)| = \frac{1}{\left|2\sin\left(\frac{\omega}{2}\right)\right| \cdot \left|\sin\left(\frac{\omega}{2}\right) + j\cos\left(\frac{\omega}{2}\right)\right|} = \frac{1}{2\sin\left(\frac{\omega}{2}\right) \cdot 1} = \frac{1}{2\sin\left(\frac{\omega}{2}\right)}$$

With 2 Integrator stages:

$$H(\omega)\big|_{\text{comb}} = \left(\frac{1}{2\sin\left(\frac{\omega}{2}\right)}\right)^2$$

In decibels:

$$dB = 20\log_{10}\left(H(\omega)\big|_{\text{comb}}\right) = 20\log_{10}\left(\frac{1}{2\sin\left(\frac{\omega}{2}\right)}\right)^2 = 40\log_{10}\left(\frac{1}{2\sin\left(\frac{\omega}{2}\right)}\right)$$

Frequency at -3 dB of the CIC filter

The $-3\,\text{dB}$ condition requires:

$$-3\,dB = 20\log_{10}(H_{\text{cic}}(\omega)) \implies \frac{H_{\text{cic}}(\omega)}{R} = \frac{10^{-3dB/20}}{R} = \frac{1}{\sqrt{2}\,R^N}$$

With 2 stages ($N = 2$), $M = 1$:

$$\frac{1}{R^2}\left(\frac{\sin\left(\frac{\omega R}{2}\right)}{\sin\left(\frac{\omega}{2}\right)}\right)^2 = \frac{1}{\sqrt{2}\,R^2}$$

For $\omega \to 0$: $\sin\left(\frac{\omega}{2}\right) \approx \frac{\omega}{2}$, so we let $\sin\left(\frac{\omega R}{2}\right) = \text{sinc}\left(\frac{\omega R}{2}\right)$:

$$\frac{1}{R^2}\left(\frac{R \cdot \sin\left(\frac{\omega R}{2}\right)}{\frac{\omega}{2} \cdot R}\right)^2 = \frac{1}{\sqrt{2}\,R^2}$$

$$\frac{1}{R^2}\,\text{sinc}^2\left(\frac{\omega R}{2}\right) = \frac{1 \cdot R}{\sqrt{2}\,R^2} \implies \text{sinc}\left(\frac{\omega R}{2}\right) = \left(R \cdot \frac{1}{\sqrt[4]{2}}\right) \approx 0{,}84 \cdot R$$

The sinc function reaches approximately 0.84 for $\frac{WR}{2} \approx 1\,\text{rad}$.

$$\frac{\omega R}{2} = 1\,\text{rad} \cdot R \implies \frac{f_{\text{out}} \cdot R}{\pi} \cdot \frac{2\pi \cdot f_{-3\text{dB}}}{2 \cdot f_{\text{out}}} = 1\,\text{rad} \cdot f_{\text{out}} \cdot \frac{1 \cdot R}{\pi}$$

$$f_{\text{out}} \cdot \frac{2\pi f_{-3\text{dB}}}{f_{\text{out}} \cdot 2} \cdot \frac{R}{\pi} = \frac{f_{out}}{\pi} = \frac{48\,\text{kHz}}{\pi} \implies f_{-3\text{dB}} \approx 15{,}2\,\text{kHz}$$

The W_{-3dB} is:

$$W_{-3dB} = 2\pi \cdot f_c = 2\pi \cdot \frac{15{,}298\,\text{kHz}}{48\,000\,\text{Hz}} \approx 1{,}99\,\text{rad}$$

Transfer function of the complete CIC filter with $N = 2$ stages

$$H_{\text{cic}}(z) = \left(\frac{1}{1 - z^{-1}}\right)^N \cdot \left(1 - z^{-RM}\right)^N \Bigg|_{N=2, M=1}$$

In the frequency domain (with $\omega = 2\pi f / f_{2{,}4\,\text{MHz}}$):

$$|H_{\text{cic}}(f)| = \left(\frac{2\sin\left(\frac{\omega MR}{2}\right)}{2\sin\left(\frac{\omega}{2}\right)}\right)^2 = \left(\frac{\sin\left(\frac{\omega_{\text{out}} R}{2}\right)}{\sin\left(\frac{\omega}{2}\right)}\right)^2 \quad \text{with } \omega = 2\pi \frac{f_{\text{in}}}{f_{2{,}4\,\text{MHz}}}$$

$$H(f) = 2\sin\left(\frac{\omega MR}{2}\right) \quad \text{(transfer function of a single COMB stage)}$$

with:

$$MR \cdot 2\pi \cdot \frac{f_{\text{null}}}{f_{\text{in}}} = K\pi \qquad f_{\text{null}} = \frac{K f_{\text{in}}}{MR} = \frac{K \cdot 2.4\,\text{MHz}}{MR}, \quad \text{with } K \in \mathbb{N}$$